

Jan. 2, 1934.

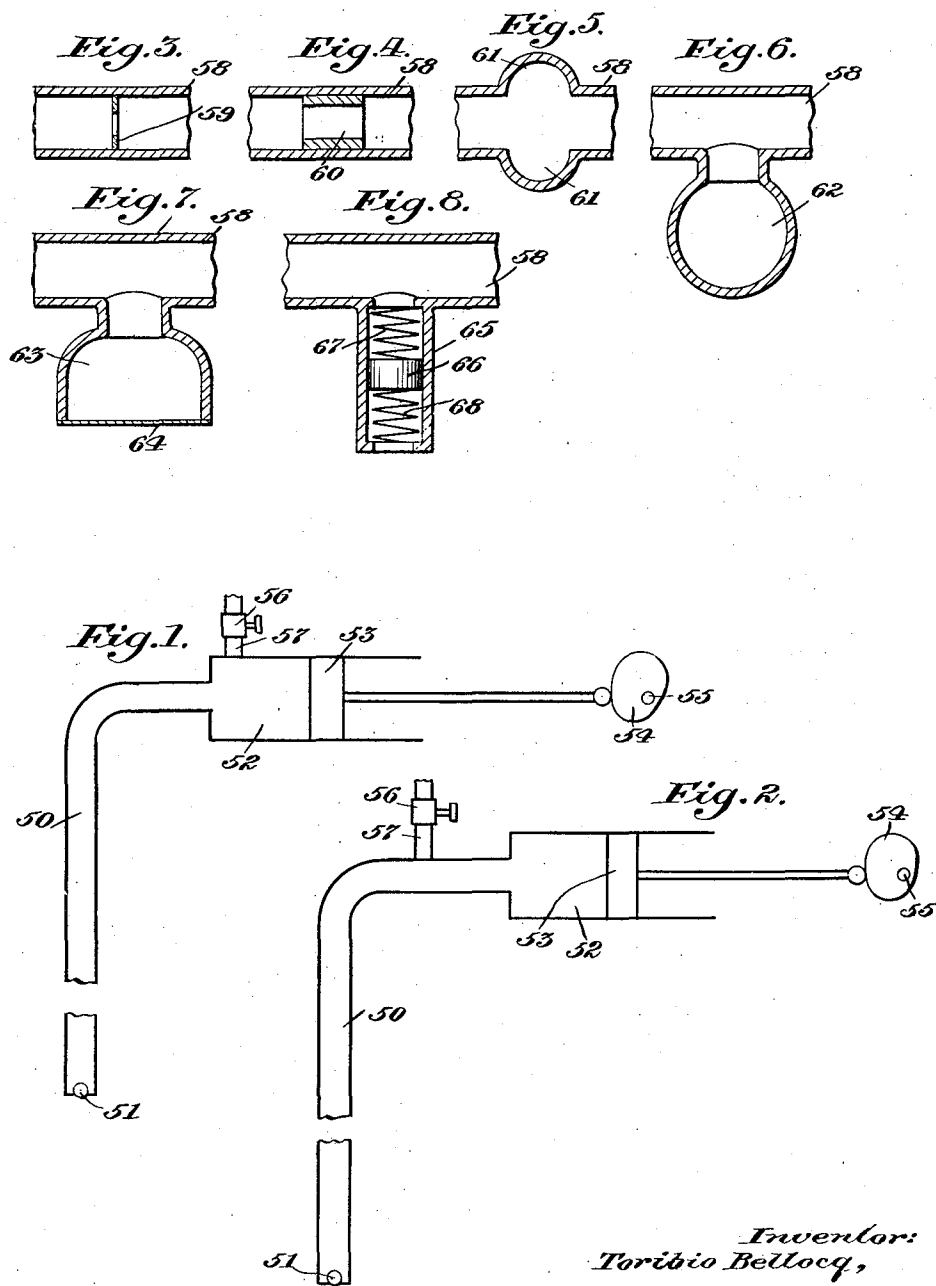
T. BELLOCQ

1,941,593

PUMPING

Filed Sept. 30, 1929

3 Sheets-Sheet 1



Inventor:
Toribio Bellocq,
by Karl Fanning
his Atty.

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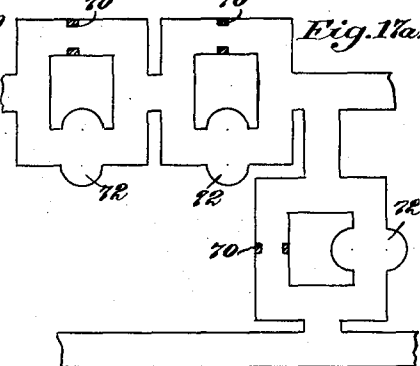
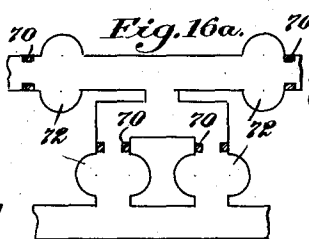
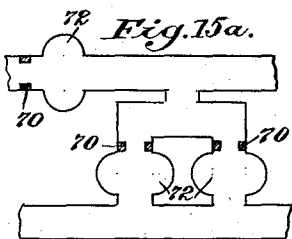
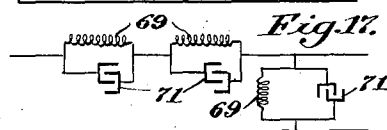
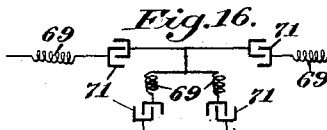
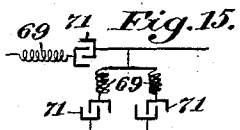
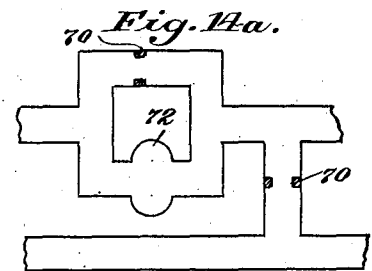
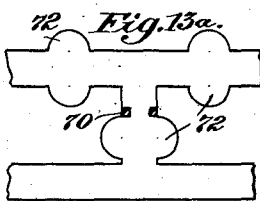
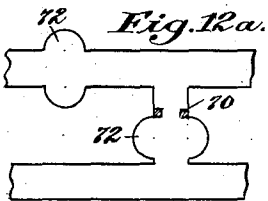
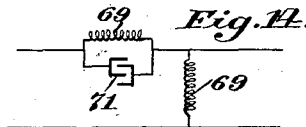
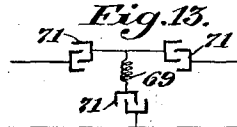
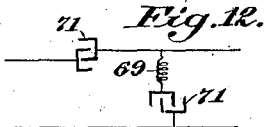
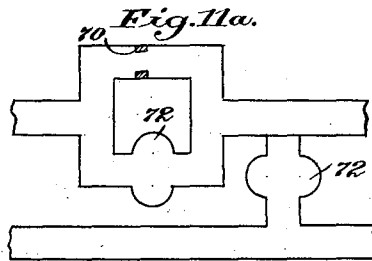
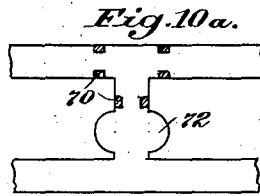
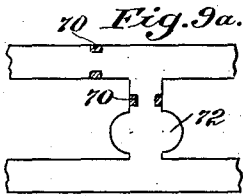
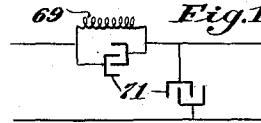
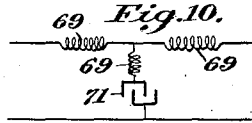
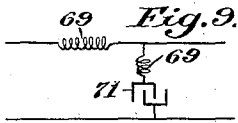
T. BELLOCQ

1,941,593

PUMPING

Filed Sept. 30, 1929

3 Sheets-Sheet 2



Inventor:
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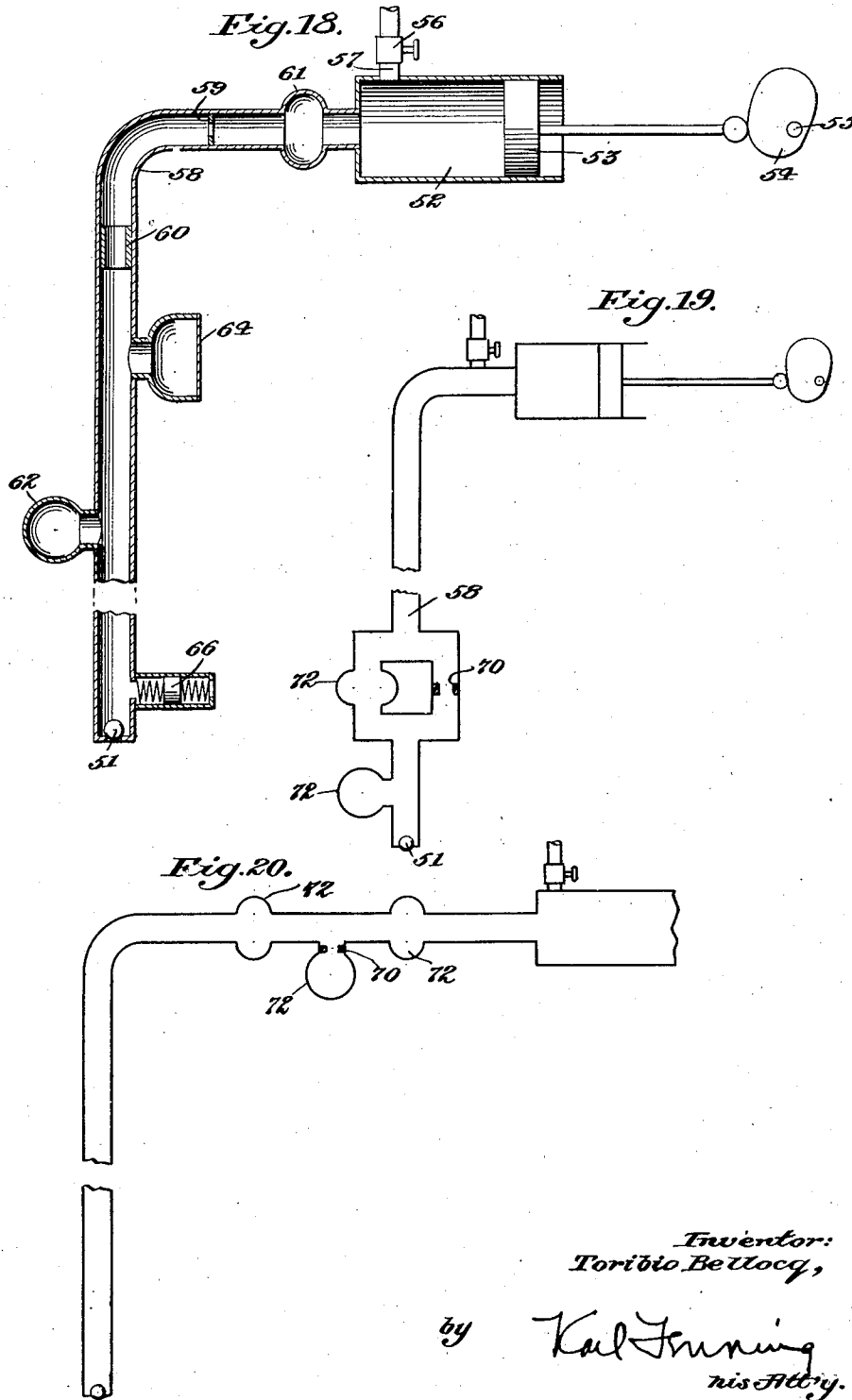
T. BELLOCQ

1,941,593

PUMPING

Filed Sept. 30, 1929

3 Sheets-Sheet 3



UNITED STATES PATENT OFFICE

1,941,593

PUMPING

Toribio Bellocq, Buenos Aires, Argentina

Application September 30, 1929

Serial No. 396,297

21 Claims. (Cl. 103—1)

The invention relates to pumping liquid especially by that method by which the energy for moving the liquid is transmitted through the liquid itself by means of compression waves or variations of compression and volume moving through the liquid whereby liquids may be extracted from substantially any depth without complicated mechanism in the well.

Pumps of this kind are described in Patent 1,730,336 and 1,730,337 issued October 1, 1929, the applications for which were pending in the Patent Office when the present application was filed.

The present invention involves the discovery that an improved pumping effect may be accomplished by modifying the relative size and arrangement of the various elements of the pumping system and by adding to or inserting therein certain auxiliary devices. These are referred to herein as filters and may be made up of inertias, capacities, resistances, leakages or any of them and may be arranged in series, in shunt or in derivation to the main liquid conveying pipe of the pumping system.

Compression waves moving in liquid are to a very considerable extent controlled by or operate in accordance with substantially the same laws as those governing alternating current electricity. The curves of pressure and flow of such waves are analogous to the voltage and amperage curves of electric current in that the difference in phase may determine the value of the wave. Inertia, capacity, resistance, or friction and leakage in a liquid system correspond more or less to induction, capacity, resistance, and losses respectively in electricity.

Variations in capacity or in leakage or in resistance or in inertia in the system effect changes therein including varying phase differences between pressure and flow of the impulses in the liquid. In an ordinary line which has its capacity, inertia, resistance and leakage more or less uniformly distributed throughout the length of the apparatus the efficiency of the line and the quantity of liquid which is to be pumped with such efficiency may be determined. Often, however, it may be desired to change the conditions of such a line to adapt it to a given speed, given efficiency or a given volume of pumped liquid. There may be produced an artificial line whose capacity, inertia, resistance and leakage will be the combination of the natural capacity, inertia, resistance and leakage with added capacity, inertia, resistance and leakage.

Inertia, as employed herein, refers to an ele-

ment in or inserted in the system for reducing the passageway at one or more points and/or decreasing the elasticity of the liquid.

Capacity, as employed herein, refers to an element in or inserted in the system for increasing the passageway at one or more points and/or increasing the elasticity of the liquid.

Leakage, as employed herein, refers to liquid leaving the system whether at the main outlet for the pumped liquid or at other points in the system some of which may be accidental, unavoidable and incalculable. The entrance of liquid into the system may be considered as negative leakage.

Resistance, as employed herein, is the retarding effect of the system on the wave movement.

Friction, as employed herein, is the resistance caused by contact of the liquid with the pipe or within the liquid.

Filter, as employed herein, refers to assemblage of inertias, capacities, resistances and leakages, or one or more of them, and this is referred to by the term "wave filter" used in the claims.

A filter may be said to be arranged in series when it is in such a portion of the system that a wave goes directly through the filter and passes on to another portion of the system. A filter may be said to be in shunt when the wave takes a divided path going through a plurality of elements of the filter. A filter may be said to be in derivation when the wave enters it from the main path of the wave and returns again to that path.

In the accompanying drawings Figure 1 is a diagrammatic illustration of one form the pumping apparatus may take. Fig. 2 is a diagrammatic illustration of another form the apparatus may take. Figs. 3 and 4 are longitudinal sections of pipes provided with inertias. Figs. 5, 6, 7 and 8 are longitudinal sections of pipes provided with capacities. Figs. 9, 10 and 11 are diagrammatic illustrations of low pass electric filters. Figs. 9a, 10a and 11a are diagrammatic illustrations of low pass acoustic filters applied to liquid systems. Figs. 12, 13 and 14 are diagrammatic illustrations of high pass electric filters. Figs. 12a, 13a and 14a are diagrammatic illustrations of high pass acoustic filters applied to liquid systems. Figs. 15, 16 and 17 are diagrammatic illustrations of band pass electric filters. Figs. 15a, 16a and 17a are diagrammatic illustrations of band pass acoustic filters applied to liquid systems.

Fig. 18 is a diagrammatic illustration of a pumping apparatus such as shown in Fig. 1 to

which is applied the devices illustrated in Figs. 3, 4, 5, 6, 7 and 8. Fig. 19 is a diagrammatic illustration of a pumping apparatus such as shown in Fig. 2 to which is applied the apparatus illustrated in Fig. 11a. Fig. 20 is a pumping apparatus such as is illustrated in Fig. 1 to which is applied the device illustrated in Fig. 13a.

The pumping system in general is illustrated as including a pipe 50 provided with a check valve 51 at its lower end intended to be inserted in the well or other source of supply of liquid to be pumped. At the other end of the pipe is shown a compression chamber or cylinder 52 in which moves a piston 53 controlled by a cam 54 on a suitably operated shaft 55. There is an outlet cock or valve 56 of any suitable type through which the liquid is pumped here shown in an outlet pipe 57. In Fig. 1 the outlet is shown leading directly from the compression chamber or cylinder 52 while in Fig. 2 it is shown as directly connected to the pipe 50 so that the outgoing liquid will not pass through the compression chamber or cylinder. The cock or valve 56 may be open at all times during operation or it may be opened and closed at appropriate times during the operation either manually or automatically.

In Fig. 3 a pipe 58 is shown provided with a narrow annular obstruction 59 which acts as an inertia for the current of waves passing through liquid in the pipe. In Fig. 4 there is shown at 58 a similar inertia 60 which takes the form of an elongated sleeve and so has a different value from the inertia shown in Fig. 3. In Fig. 5 is shown a pipe 58 provided with rigid protuberances 61 which act as a capacity in series with the pipe. In Fig. 6 is shown a pipe 58 which is connected to a closed rigid chamber 62 which when filled with liquid acts as a capacity in derivation. In Fig. 7 is shown a pipe 58 connected with a chamber 63 closed at its outer end by a diaphragm 64. When the chamber 63 is filled with liquid it acts as a capacity in derivation. In Fig. 8 is shown a pipe 58 from which leads a pipe 65 in which is a snugly fitting piston 66 movable against springs 67 and 68. The resiliency of the springs takes the place of liquid resiliency and acts as a capacity in derivation to the pipe 58. These are illustrative only and other forms of inertia and capacity may be employed.

The pipe through which liquid is to be pumped may be horizontal, vertical or inclined or bent. If there is a bend in the pipe the relation between the parts on each side of the bend tends to produce an harmonic of the principal wave and in order to work most efficiently this harmonic should preferably have a node in the elbow at the bend. The elbow thus acts as an inertia.

The entrance or the exit of the liquid pumped may also tend to produce an harmonic four times as long as the length of the liquid which passes through the pipe at each stroke of the compressing device. It is desirable to have this harmonic the same as the harmonic produced by the proportion between the portions of a bent line, or at least an harmonic thereof. The flowing of the liquid through the inlet or the outlet has somewhat the effect of an inertia. The change in size between the pipe carrying the liquid and the cylinder of the compressor at their point of connection has somewhat the same effect as an inertia. The small chamber generally employed at the outlet cock or valve between the com-

pressor piston and the pipe line and likewise the cylinder itself each have somewhat the same effect as a capacity. This chamber may be omitted and the cylinder alone be referred to as the compression chamber.

There are several elements of the apparatus which may be adjusted to change or vary the efficiency of operation including: (1) the size of the compression chamber may be varied; (2) the relative size of the connection between the compression chamber and the pipe may be varied; (3) the cross-section of the pipe through which the liquid is pumped may be varied; (4) the speed or timing of the piston may be varied; (5) the size or cross-section of the piston or its length of stroke may be varied (6) the size or opening of the check valve through which liquid enters the system may be varied; or (7) the size and suddenness of the opening of the outlet may be varied. In order to procure proper and appropriate adjustments or regulation of the system any or all of the various elements may be appropriately and correspondingly varied. The quantitative effect of variations in the different elements of the apparatus is not the same.

A difference in operation is produced when the outlet is in the compression chamber from that produced when the outlet is on the pipe. In the first arrangement increasing the size or opening of the check valve or in the second arrangement decreasing the size or the opening of the check valve effects a variation in the same direction as (1) increasing the size of the compression chamber (2) decreasing the relative size of the connection between the chamber and the pipe (3) decreasing the cross-section of the pipe through which the liquid is pumped (4) increasing the speed or decreasing the timing of the piston (5) increasing the size or cross-section of the piston or its length of stroke (6) increasing the size of opening of the outlet. This indicates the direction of change to be made to compensate for any change in the apparatus. If the outlet on the pipe is removed farther from the connection between the pipe and the compression chamber or if inertia associated with the outlet is moved farther from the connection between the pipe and compression chamber the effect is of the same character as would be an enlargement of the compression chamber. It is generally desirable, however, to put the outlet, the inertia and the connection between the pipe and the compression chamber as close together as possible because when they are separated one from another there is no longer merely the effect of a chamber assembled at one place but the effect of a chamber like a piece of pipe with different inductions in each extremity. With such an arrangement it is more difficult to calculate the exact value of such a chamber. Sometimes, however, it is found desirable to use such an extended chamber despite the difficulty in calculating its characteristics.

It may happen that when the pipe is very long the wave will be deformed as it proceeds through the liquid. It may then be desirable to put additional or new capacities, inertias, resistance or leakage in one or several parts of the apparatus to prevent this deformation of the current. Somewhat similarly in telegraphic and telephonic lines traveling great distances means are provided to prevent deformation of the circuit wave such as are provided by the system of Puppini or others.

It may be desirable to choose a vibration or

5 wave of such a character that the liquid will
vibrate with the natural vibration produced by
gravity. For example a compressor may be so
operated as to cause an artificial acceleration
10 during its stroke corresponding to the natural
acceleration produced by gravity on the liquid
freely falling in the system. When this natural
vibration is not convenient for the quantity of
liquid it is desired to pump the constants of the
15 line may be appropriately changed to produce the
desired effect.

When the pipe is large enough or the quantity
of the liquid pumped is small in relation to the
size of the pipe, the resistance will be small and
for practical purposes may be neglected. It is
desirable that the leakage in the system, be re-
duced to a minimum so that the only value of
leakage to be considered may be that produced
by the liquid pumped through the outlet. En-
20 trance of liquid into the system may be con-
sidered as negative leakage.

The compression waves in the liquid of the
pumping system are somewhat analogous to sound
waves. Therefore, elements which modify the
25 waves in the liquid of the pump may for con-
venience be referred to as acoustic elements and
it is to be understood that when the term acoustic
or hydraulic is employed herein it relates to an
element modifying the compression waves in the
30 liquid more or less in accordance with the laws
of sound waves in the liquid. This may be a con-
venient form of expression and it may aid in un-
derstanding the present invention to bear in mind
the analogy between the compression waves in
35 the liquid in the pumping system and sound
waves.

Acoustic inertia or self induction may be an
apparatus having only inertia which may be
formed for instance by a reduction of a section
40 of the pipe. Likewise acoustic capacity may be
a chamber communicating with the pipe and
filled with the liquid. Acoustic inertias, capaci-
ties, resistances and leakages may be combined
to form acoustic filters analogous to electric fil-
45 ters and their effect on the compression waves
in the pumping system may be analogous to
their effect on sound waves. As in electricity
such acoustic filters in the pumping system may
be referred to as low pass filters, high pass fil-
50 ters and band pass filters. The low pass filters
may let pass or produce vibrations of a fre-
quency lower than a certain speed or may so
convert other vibrations received. The high
pass filters may let pass or may produce vibra-
55 tions of a frequency higher than a certain speed
or may so convert other vibrations received. The
band pass filters may let pass or may produce
vibrations of a frequency between two certain
speeds, or may so convert other vibrations re-
60 ceived. All of these filters may stop or not let
pass other vibrations. Acoustic filters may be
arranged in accordance with the same laws as
electric filters. It may be easier to make filters
occupying a very short space in relation to the
65 wave length. It is possible, however, to employ
other longer filters but the exact determination
of their effect may be more difficult.

The analogies between electric filters and
acoustic filters exist not only in words but they
70 are true physical analogies. The phenomena of
the propagation of an electric current and of
sound waves and of compression waves in liquid
are controlled by the same laws. In dealing
with these waves the combination of acoustic
75 inertia, capacities, resistance and leakages ac-

complish similar results as are accomplished by
combinations of similar elements in electric cir-
cuits. In order to facilitate determining the
effect of a filter it is desirable that the two ex-
80 tremities of the filter be of the same value. It
is possible, however, to employ filters having ex-
tremities of different values. In such an ar-
rangement the difference in the two extremi-
ties may produce an effect of itself. The ideal
filter is one which without substantial loss of
85 energy will let pass the frequency which it is
desired to employ in the apparatus and at the
same time stop other frequencies. Generally
filters of this character cannot be completely
realized in practice, but it is desirable to employ
90 filters approaching this ideal as nearly as possi-
ble. A general filter may be employed or sev-
eral filters may be employed in a single system
being associated or combined with similar or
different filters. 95

Generally electric circuits employ two wires
or lines and the induction, capacity, resistance
and leakage may be installed in one or both or
between them in series, derivation, or in shunt.
In liquid systems, especially pumping systems, 100
usually there is but a single line or pipe but the
inertia, capacity, resistance and leakage may be
placed in series or in shunt on the line or in de-
rivation on the line in which case the effect is
as if the element in derivation communicated at 105
one side with the line and at the other side with
a ground analogous to the grounding of an elec-
trical device.

The effect of filters put in derivation may be
varied, for example by putting a capacity in de-
110 rivation made not of a closed chamber filled
with the same liquid but of a closed chamber
filled with another liquid of different density or
elasticity, separated by a membrane from the
other liquid or by a chamber filled by the same 115
liquid but closed by a membrane which sepa-
rates the liquid from the air. The capacity thus
is to be considered as not between the line and
the ground but between the line and another
artificial ground. The natural vibration of the 120
liquid will be different in the different cases.
This chamber filled with the different liquid or
provided with a membrane may be replaced by
a cylinder having a floating piston maintained
by springs which make the piston work like a 125
membrane. The introduction of any of these
different apparatus having vibration character-
istics different from the natural vibrations of the
line has the effect of changing the constants of
the line and may be employed in a manner simi- 130
lar to the more usual type of filters. Any other
suitable form of capacity, inertia, resistance or
leakage may be employed.

For installations operated under substantially
constant conditions a filter of exact characteris- 135
tics may be employed, but for other installations
less exact filters may be employed.

In Figs. 9, 10 and 11 are shown examples of
characteristic electrical low pass filters. In Figs.
9a, 10a and 11a are shown similar characteristic 140
low pass acoustic filters applied to liquid sys-
tems. The arrangement in Fig. 9a corresponds
to that in Fig. 9; the arrangement in Fig. 10a
corresponds to that in Fig. 10, while that in Fig.
11a corresponds with that in Fig. 11. In each 145
instance the electrical inertias 69 are replaced
by acoustic inertias 70 while the electric capaci-
ties 71 are replaced by acoustic capacities 72.
Figs. 12, 13 and 14 illustrate types of electric high
pass filters and Figs. 12a, 13a and 14a illustrate 150

types of acoustic high pass filters corresponding respectively to the electric filters illustrated in Figs. 12, 13 and 14. Figs. 15, 16, and 17 illustrate diagrammatically electric band pass filters and
 5 Figs. 15a, 16a and 17a illustrate characteristic acoustic band pass filters applied to liquid systems corresponding respectively to the filters illustrated in Figs. 15, 16 and 17. In all of these figures the electrical inertias 69 correspond to
 10 the acoustic inertias 70 and the electric capacities 71 correspond to the acoustic capacities 72. The filter arrangements here illustrated are selected as types only. These and other appropriate electric filters may be followed with like
 15 effect and the corresponding acoustic filters inserted at appropriate points in liquid systems with corresponding results and the term filter is used to include all such devices as rectify, transform, alter, or modulate the currents of the compression waves set up in the liquid in the pumping
 20 system.

In Fig. 18 is shown a simple pumping system such as shown in Fig. 1 but modified by being associated with the filters or wave modifying devices illustrated in Figs. 3 to 8 inclusive. The
 25 capacities 61, 62, 64 and that having the piston 66 are associated with inertias 59 and 60. As illustrated the inertia 59 and the capacity 61 are shown relatively near to the pump cylinder 52 and the other wave modifying devices are distributed throughout the line 66 being very near
 30 to the inlet valve 51. 59, 60 and 61 are in series while 62, 64 and 66 are in derivation. This specific arrangement is not essential nor is it essential that all of these devices be incorporated in a single system. In general a single device may be
 35 sufficient. For instance the inertia 59 might be employed and the devices 60, 61, 62, 64 and 66 omitted or any two or more of them might be omitted. Likewise the particular position of these various devices in the system is not essential. For instance the device 66 which is shown as
 40 near the check valve 51 might be installed where the device 61 is illustrated near the pump cylinder 52 or any other desired arrangement and location of any or all of these devices may be made. Instead of assembling in a single system a plurality of different wave modifying devices as
 45 illustrated in Fig. 18 a plurality of similar devices might be employed where more than one is desired and the location of all of the devices is interchangeable. When apparatus such as is illustrated in Fig. 1 is operated the wave produced in the liquid will have certain characteristics and
 50 for the conditions of operation it may be that that wave is satisfactory and efficient. It may be, however, that for the conditions under which the apparatus is operating the wave produced by apparatus such as illustrated in Fig. 1 will
 55 not have the desired characteristics and may not be effectively efficient to produce the desired pumping effect. By adding to the apparatus shown in Fig. 1 any one or more of the devices illustrated in Figs. 3 to 17 the characteristic of
 60 the wave may be modified so as to produce the desired effective efficient result. Thus the wave set up in such an apparatus as Fig. 18 will have characteristics different from the wave set up in such an apparatus as Fig. 1 and the addition or
 70 removal of any one of the impedance or capacity devices illustrated in Fig. 18 will correspondingly modify the wave.

In Fig. 19 is shown assembled in such a pump as is shown in Fig. 2 a filter such as illustrated
 75 in Fig. 11a and in Fig. 20 such a filter as illus-

trated in Fig. 13a is illustrated in connection with such a pump as shown in Fig. 1. As indicated, when the arrangement is applied to a pumping system there may be present only one pipe line
 80 the other pipe being omitted and replaced by an artificial ground similarly to the arrangement by which a two wire electric system may be employed with only one wire and a ground. In the pumping system the effect of a ground may be
 85 procured by closing the connection to the omitted pipe. Likewise the filters illustrated in Figs. 9a, 10a, 12a, 14a, 15a, 16a and 17a or any grouping or combinations of them or other filters may be assembled into a single apparatus. The arrangement illustrated in Fig. 19 is in part a
 90 shunt arrangement and, incorporating a low-pass filter, will modify the waves in the pumping system and allow only those waves having frequencies lower than a certain desired speed or it may convert to such speed other vibrations in the system. Similarly the filter associated with
 95 the pumping system in Fig. 20 being a highpass filter will modify the waves in the pumping system and allow only those waves having frequencies higher than a certain desired speed or it may
 100 convert to such speed other vibrations in the system.

The devices shown are only preferred embodiments of the invention, but any other suitable devices may be used for the same purposes.
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The weight of the check valve or the strength of springs sometimes controlling it which change the character of its opening change the value of the inertia produced by its opening. The check valve being inertia and so a filter element may
 110 be omitted and the other elements of the apparatus adjusted accordingly. When more than one pump is in the pipe line each pump may act as check valve or inlet for the succeeding pump.

Variations in the speed of the piston during a
 115 stroke are of importance. When the piston is moved by a crank shaft the speed increases from zero to maximum speed during the first half of the time of the entrance of the piston (that is, the first quarter of a complete cycle) and produces an acceleration in the lifting of the liquid. In the second half of the time of entrance of the piston (that is, the second quarter of a complete cycle) the speed of the piston decelerates going
 120 from its high speed to zero speed. Thus the liquid tends to be lifted with more strength when the piston is entering during the first half of its forward stroke and tends, because of the deceleration, to be lifted with less strength in each unit of time when the piston is entering during the second half of its forward stroke. In general under these conditions it is better to open the outlet valve when or before the piston reaches the half point of the inward stroke. Sometimes it may be better to open the valve at the end of
 125 the entrance stroke of the piston or somewhat before. When a proper cam is employed to operate the piston, the piston speed increases throughout the entire advance movement so the same effect may sometimes be produced by a cam which gives only one-half the length of movement to the piston as would be given by a corresponding crank. When such a cam is used only
 130 only-half as much liquid may be forced out of the piston cylinder as with a crank and hence at each stroke less liquid need go into the cylinder to keep it full in order to get as much liquid output from the apparatus. With the most efficient adjustment of the apparatus, during the acceleration of the piston liquid is lifted, that
 135
 140
 145
 150

is to say, the liquid may be moving from the check valve to the compressor. If the valve is opened at this moment the liquid which is already moving upward, will tend to be lifted with more efficiency since it is not obliged to change its direction.

The suddenness of opening the outlet being the principal element in the transformation of the compression waves should be especially considered and arranged to cooperate with the rest of the system.

In certain cases it may be desired to produce a certain pressure in the apparatus at the time the outlet is opened as in this connection it seems that the liquid pumped going out with a certain pressure will carry with it enough energy to lift the liquid higher than the apparatus. The apparatus working in this form may work somewhat like a combination of the system employing the open cock at the outlet and the system opening and closing the valve, retaining thus the advantages of both systems.

The conditions involved in the use and application of filters indicate the important effect which may be produced by small defects in the apparatus such as leakage, holes or couplings or connections which allow liquid to pass out, these defects being analogous to the bad effect produced in electrical circuits by bad contacts or by other defects in different parts of the apparatus, etc. The piston may work either with the cylinder horizontal or vertical or inclined, or any other suitable means may be employed to produce variations in compression in the liquid to set up the compression waves. The pipe through which liquid flows may be either horizontal or vertical or inclined, and it may be curved, bent or straight. All of these arrangements may be referred to as pumps, and the term pumping is used as including moving, transporting or conveying in vertical or horizontal or inclined directions or a combination of any of them, and the liquids so pumped may be simple liquids or may carry more or less solid matter or gases.

I claim as my invention:

1. The method of varying and adjusting the efficiency of pumping apparatus comprising a pipe carrying liquid an inlet and an outlet therefor and means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, comprising associating and communicating a wave filter with the pipe so as to modify the waves in the liquid.

2. Pumping apparatus comprising a pipe carrying liquid, an inlet and an outlet therefor, means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, a wave filter associated and communicating with the pipe so as to modify the waves in the liquid.

3. Pumping apparatus comprising a pipe carrying liquid, an inlet and an outlet therefor, means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, inertia, capacity and leakage associated and communicating with the pipe so as to modify the waves in the liquid.

4. Pumping apparatus comprising a pipe carrying liquid, an inlet and an outlet therefor, means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, inertia associated and communicating with the pipe so as to modify the waves in the liquid.

5. Pumping apparatus comprising a pipe car-

rying liquid, an inlet and an outlet therefor, means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, capacity associated and communicating with the pipe so as to modify the waves in the liquid.

6. Pumping apparatus comprising a pipe carrying liquid, an inlet and an outlet therefor, means for causing compression waves to pass through the liquid and thereby cause movement of liquid through the apparatus, inertia and capacity associated and communicating with the pipe so as to modify the waves in the liquid.

7. Pumping apparatus comprising a pipe carrying liquid, means for causing compression waves to pass through the liquid, whereby the waves cause liquid to flow through the apparatus, and an auxiliary wave filter associated and communicating with the pipe so as to modify the waves.

8. Apparatus for pumping liquids comprising a check valve in the liquid supply, a compressor at the point of delivery including a cylinder and a piston and an outlet pipe, a pipe filled with liquid extending from the valve to the compressor, means for reciprocating the piston rapidly through a short stroke whereby the compression of the liquid produced by the piston is varied and a wave filter associated and communicating with the last named pipe which when altered varies the output of the pump.

9. An apparatus for pumping liquid comprising a pipe filled with liquid leading to the liquid supply, a check valve in the pipe in the liquid supply, a compressor cylinder at the other end of the pipe, an open outlet from the cylinder, a piston to alternately compress the liquid in the pipe and release the pressure, and a wave filter associated and communicating with the pipe.

10. Apparatus for pumping liquid comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure to set up waves in the liquid, an outlet for the pipe whereby the waves cause liquid to enter and flow through the pipe and auxiliary wave filters associated and communicating with the pipe which when altered vary the liquid flow through the pipe.

11. Apparatus for pumping liquid comprising means for alternately compressing a body of liquid in a pipe, separate means for alternately releasing the pressure, an outlet for the pipe whereby the variations in compression cause the liquid to enter and flow through the pipe, and an auxiliary wave filter for varying the flow.

12. Apparatus for pumping liquids comprising a pipe leading to the liquid supply filled with liquid, a check valve in the pipe within the liquid to be moved, an outlet for the liquid, means for compressing the liquid in the pipe, separate means for suddenly releasing the pressure, and an auxiliary wave filter for varying the flow of the liquid.

13. Apparatus for pumping liquids comprising a pipe leading to the liquid supply and filled with liquid, a check valve in the pipe within the liquid supply, an outlet for the liquid, means to set up a contemporaneous plurality of series of compression waves in the liquid in the pipe to operate the check valve and move the liquid, and an auxiliary wave filter for varying the flow of the liquid.

14. Apparatus for pumping liquids comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure, an outlet for the pipe, whereby the variations in

compression cause liquid to enter and flow through the pipe, and an enlargement connected with the pipe modifying the pumping effect.

15. Apparatus for pumping liquids comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure, an outlet for the pipe, whereby the variations in compression cause liquid to enter and flow through the pipe, and an enlargement in the pipe modifying the pumping effect.

16. Apparatus for pumping liquids comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure, an outlet for the pipe, whereby the variations in compression cause liquid to enter and flow through the pipe, and an obstruction in the pipe modifying the pumping effect.

17. Apparatus for pumping liquids comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure, an outlet for the pipe, whereby the variations in compression cause liquid to enter and flow through the pipe, and an obstruction and an enlargement in the pipe modifying the pumping effect.

18. Apparatus for pumping liquids comprising means for alternately compressing a body of liquid in a pipe and releasing the pressure, an outlet for the pipe, whereby the variations in compression cause liquid to enter and flow through the pipe, and an obstruction and an enlargement in a passageway connected with the pipe modifying the pumping effect.

19. The method of regulating the output of a pumping apparatus comprising means for alternately compressing a body of liquid in a pipe, separate means for alternately releasing the pressure and an outlet for the pipe, whereby the variations in compression cause the liquid to enter and flow through the pipe, consisting of inserting and regulating, varying and/or adjusting auxiliary and additional wave filter elements.

20. The method of regulating the output of apparatus for pumping liquid comprising means for alternately compressing a body of liquid in a pipe, separate means for alternately releasing the pressure, an outlet for the pipe whereby the variations in compression cause the liquid to enter and flow through the pipe, and an auxiliary wave filter for varying the flow, consisting of regulating, varying and/or adjusting the auxiliary wave filter.

21. The method of regulating the output of apparatus for pumping liquids comprising a pipe leading to the liquid supply and filled with liquid, a check valve in the pipe within the liquid supply, an outlet for the liquid, means to set up a contemporaneous plurality of series of compression waves in the liquid in the pipe to operate the check valve and move the liquid, and an auxiliary wave filter for varying the flow of the liquid, consisting of regulating, varying and/or adjusting the auxiliary wave filter.

TORIBIO BELLOCQ.

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